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# **1/5" 5Mega CMOS Image Sensor**

## **BF2557CS**

### **芯片应用和设计指导**



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# 1 外围电路

## 1.1 原理图

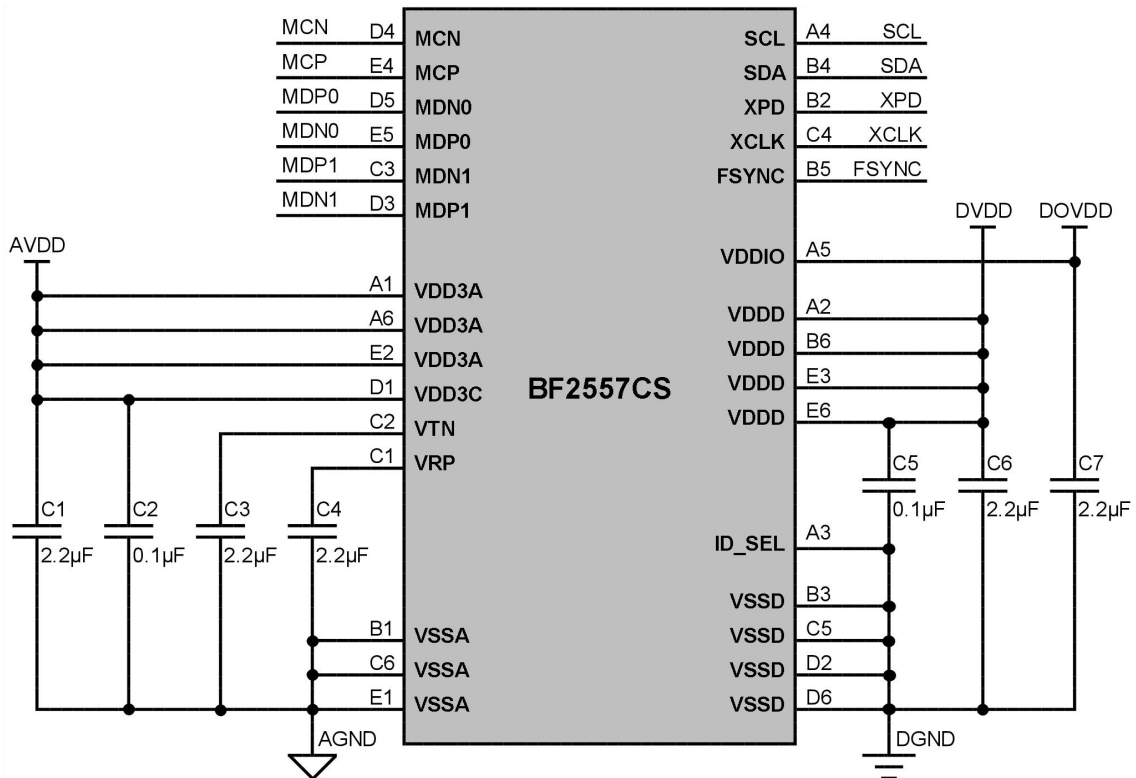


图 1-1 原理图

## 1.2 外围电路设计说明

- BF2557CS 芯片分四路电源供电：
  - VDDIO, 1.7V ~ 3.1V
  - VDD3A, 2.6V ~ 3.1V
  - VDD3C, 2.6V ~ 3.1V
  - VDDD, 1.15V ~ 1.35V
- 建议所有电源和地线宽度 0.2mm 以上，电源走线需先经过滤波电容再进入 sensor 的电源管脚；
- AGND 与 DGND 分开走线；
- I2C 地址选择：当 ID-SEL 者接 DGND 时，I2C 地址为 0x7c(W)/0x7d(R)，ID-SEL 接 DOVDD 时，I2C 地址为 0xdc(W)/0xdd(R)；
- MCP/MCN、MDP0/MDN0 和 MDP1/MDN1 为三对差分信号，每对差分信号走线尽可能不要换层打孔，走线时每对差分信号同层走线、两侧需用 GND 包覆屏蔽，背层尽量铺地作为参考层；(Differential impedance=100Ω)

- 由于芯片封装锡球排布影响,封装上 MCN 比 MCP 长 0.5mm,MDN0 比 MDP0 长 0.5mm,MDN1 比 MDP1 长 0.5mm,建议在 Layout 布线时 MCN 比 MCP 短 0.5mm,MDN0 比 MDP0 短 0.5mm,MDN1 比 MDP1 短 0.5mm,达到整体等长。尽量确保线对内整体长度差异小于 0.1mm,线对之间整体长度差异小于 0.2mm。

## 2 管脚配置

### 2.1 管脚示意图

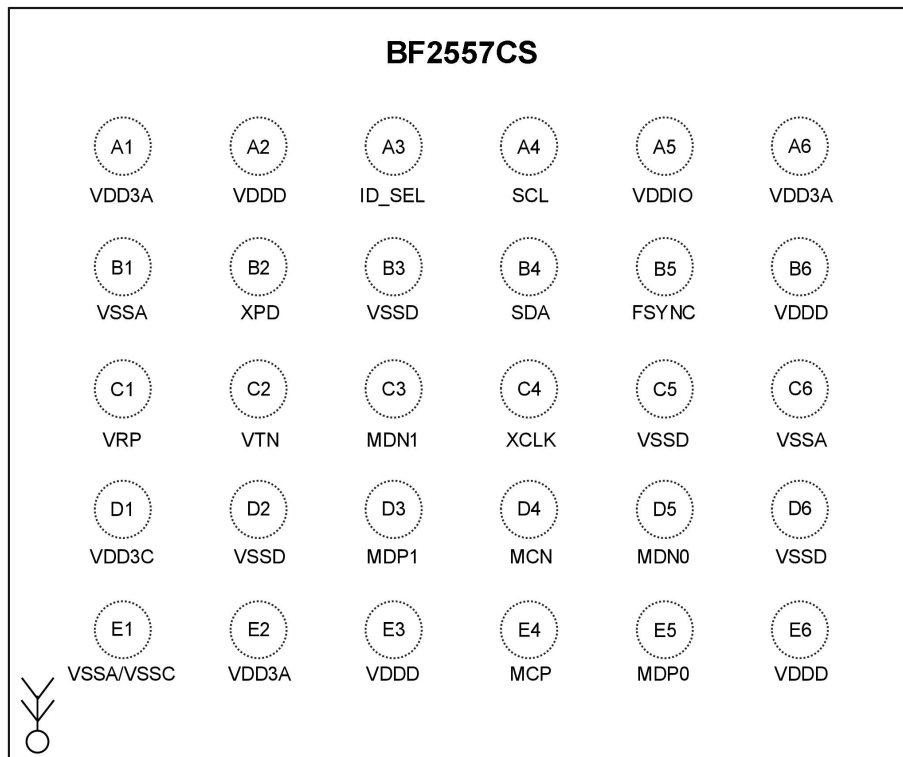


图 2-1 管脚阵列（顶视图）

### 2.2 管脚说明

表 2-1 管脚说明

| Pin Number | Name   | Pin Type | Function/Description                                                     |
|------------|--------|----------|--------------------------------------------------------------------------|
| A1         | VDD3A  | Supply   | Analog power supply: 2.6V~3.1V                                           |
| A2         | VDDD   | Supply   | Digital core power supply: 1.15V~1.35V                                   |
| A3         | ID_SEL | Input    | I2C Address Selection.<br>ID_SEL=Low: 0x7c/0x7d; ID_SEL=High: 0xdc/0xdd. |
| A4         | SCL    | Input    | SCCB serial interface clock input                                        |
| A5         | VDDIO  | Supply   | I/O power supply: 1.7V~3.1V                                              |
| A6         | VDD3A  | Supply   | Analog power supply: 2.6V~3.1V                                           |
| B1         | VSSA   | Ground   | Analog ground                                                            |
| B2         | XPD    | Input    | Power Down Mode Selection.                                               |



|    |       |        |                                        |
|----|-------|--------|----------------------------------------|
|    |       |        | 0: Power down mode; 1: Normal mode.    |
| B3 | VSSD  | Ground | Digital ground                         |
| B4 | SDA   | I/O    | SCCB serial interface data inout       |
| B5 | FSYNC | I/O    | Frame Sync                             |
| B6 | VDDD  | Supply | Digital core power supply: 1.15V~1.35V |
| C1 | VRP   | I/O    | Inner analog voltage.                  |
| C2 | VTN   | I/O    | Inner analog voltage.                  |
| C3 | MDN1  | I/O    | Neg Data lane1 of MIPI                 |
| C4 | XCLK  | Input  | System input clock                     |
| C5 | VSSD  | Ground | Digital ground                         |
| C6 | VSSA  | Ground | Analog ground                          |
| D1 | VDD3C | Supply | Analog power supply: 2.6V~3.1V         |
| D2 | VSSD  | Ground | Digital ground                         |
| D3 | MDP1  | I/O    | Pos Data lane1 of MIPI                 |
| D4 | MCN   | I/O    | Neg Clock lane of MIPI                 |
| D5 | MDN0  | I/O    | Neg Data lane0 of MIPI                 |
| D6 | VSSD  | Ground | Digital ground                         |
| E1 | VSSA  | Ground | Analog ground                          |
| E2 | VDD3A | Supply | Analog power supply: 2.6V~3.1V         |
| E3 | VDDD  | Supply | Digital core power supply: 1.15V~1.35V |
| E4 | MCP   | I/O    | Pos Clock lane of MIPI                 |
| E5 | MDP0  | I/O    | Pos Data lane0 of MIPI                 |
| E6 | VDDD  | Supply | Digital core power supply: 1.15V~1.35V |

## 3 结构规格

### 3.1 结构图

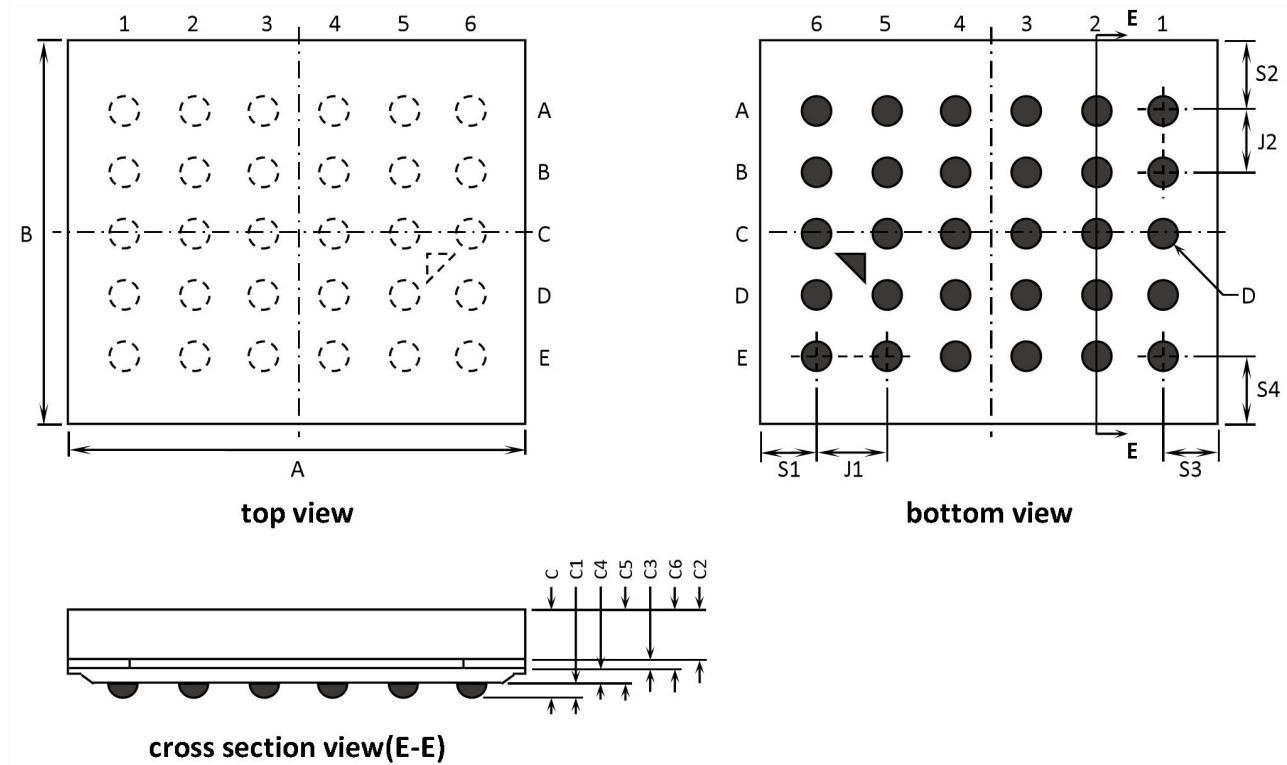


图 3-1 封装规格图

### 3.2 封装尺寸

表 3-1 封装尺寸

| Description                               | Symbol | Nominal     | Min   | Max   |
|-------------------------------------------|--------|-------------|-------|-------|
|                                           |        | Millimeters |       |       |
| Package Body Dimension X                  | A      | 3.730       | 3.705 | 3.755 |
| Package Body Dimension Y                  | B      | 3.120       | 3.095 | 3.145 |
| Package Height                            | C      | 0.620       | 0.565 | 0.675 |
| Ball Height                               | C1     | 0.120       | 0.090 | 0.150 |
| Glass Thickness                           | C2     | 0.300       | 0.290 | 0.310 |
| Cavity wall height                        | C3     | 0.035       | 0.031 | 0.039 |
| Si Thickness                              | C4     | 0.130       | 0.120 | 0.140 |
| Package Body Thickness                    | C5     | 0.500       | 0.465 | 0.535 |
| Thickness from top glass surface to wafer | C6     | 0.335       | 0.330 | 0.340 |
| Ball Diameter                             | D      | 0.230       | 0.200 | 0.260 |
| Total Ball Count                          | N      | 30          |       |       |

|                                      |    |       |       |       |
|--------------------------------------|----|-------|-------|-------|
| Pins pitch X axis                    | J1 | 0.560 | 0.550 | 0.570 |
| Pins pitch Y axis                    | J2 | 0.500 | 0.490 | 0.510 |
| Edge to Ball Center Distance along X | S1 | 0.465 | 0.435 | 0.495 |
| Edge to Ball Center Distance along Y | S2 | 0.565 | 0.535 | 0.595 |
| Edge to Ball Center Distance along X | S3 | 0.465 | 0.435 | 0.495 |
| Edge to Ball Center Distance along Y | S4 | 0.555 | 0.525 | 0.585 |

## 4 光学中心

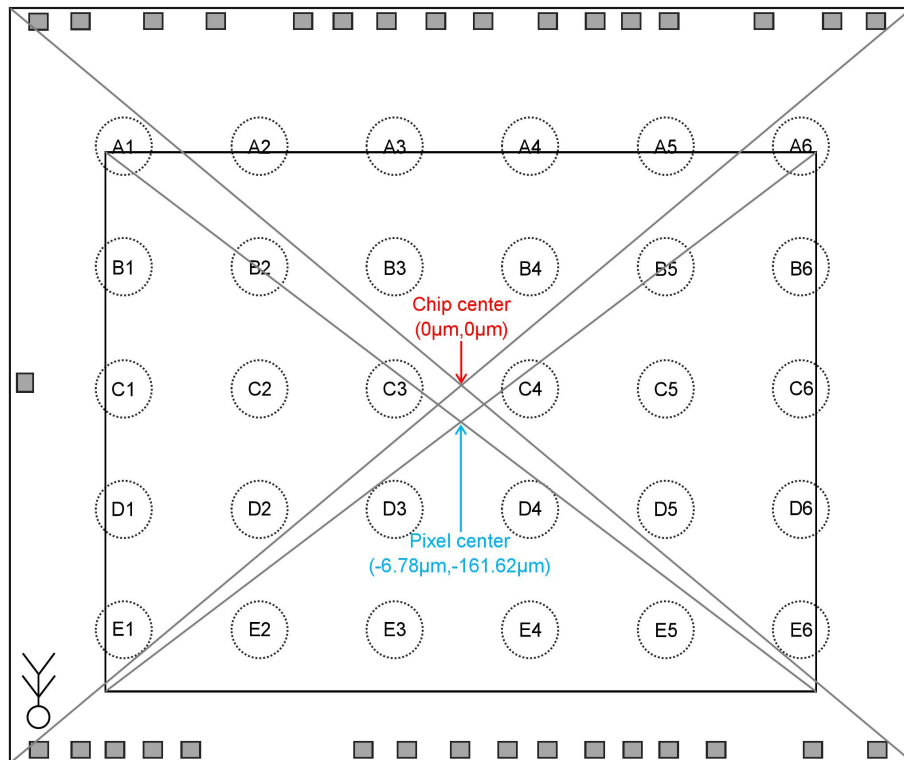


图 3-1 光学中心（顶视图）

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